8:- Find the value of $\sin 45^\circ$, $\cos 45^\circ$, $\tan 45^\circ$, $\operatorname{cosec} 45^\circ$, $\sec 45^\circ$ and $\cot 45^\circ$

ans:- In a right rectangle with one acute angle 45°, the ~~order~~ other acute angle is also 45°. Hence the two legs are equal

let each leg be a. Pythagorus.
hypotenuse $= a\sqrt{2}$

Therefore:

$\sin 45^\circ = a/(a\sqrt{2}) = 1/\sqrt{2}$

$\cos 45^\circ = a/(a\sqrt{2}) = 1/\sqrt{2}$

$\tan 45^\circ = a/a = 1$

$\operatorname{cosec} 45^\circ = \sqrt{2}$

$\sec 45^\circ = \sqrt{2}$

$\cot 45^\circ = 1$